

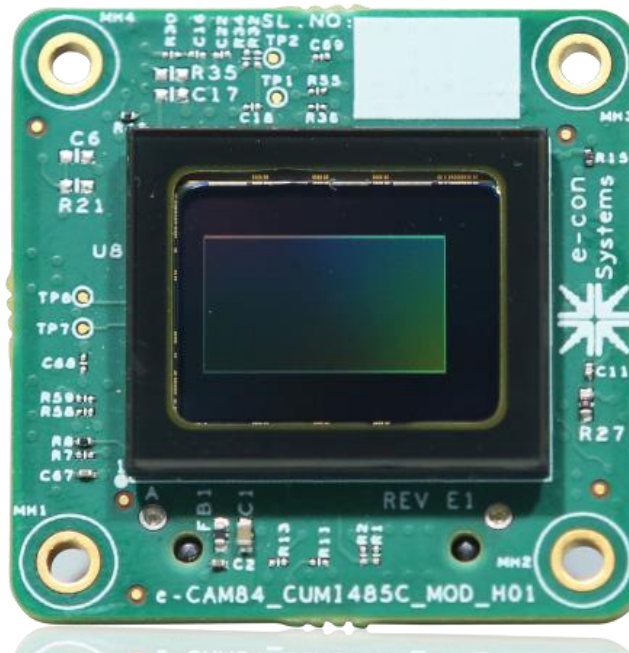


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e-CAM84_CUMI485C_MOD_H02R1



Camera Module Datasheet

Revision 1.3
14th Mar 2023



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1 Revision History

Rev	Date	Description	Author
1.0	03-June-2022	Initial Draft	Camera Hardware Team - Products
1.1	16-Nov-2022	Added Xavier support	Camera Hardware Team - Products
1.2	22-Feb-2023	4-Lane support	Camera Hardware Team - Products
1.3	14-Mar-2023	Added 2-lane support	Camera Hardware Team - Products



2 Introduction

e-CAM84_CUMI485C_MOD_H02R1 is a high performance, 8.4 MP auto low light camera module. It is based on IMX485 sensor from STARVIS family of CMOS image sensor from SONY. e-CAM84_CUMI485C_MOD_H02R1 is designed to connect with any application processor that has MIPI interface.

This document serves as the datasheet for e-CAM84_CUMI485C_MOD_H02R1 with electrical, mechanical and software features.

3 Disclaimer

The specifications and features of e-CAM84_CUMI485C_MOD_H02R1 camera board are provided here as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

4 Description

e-CAM84_CUMI485C_MOD_H02R1 manual focus camera module has good low light performance. It supports 4K at 60 fps for maximum video performance.

The camera module is based on IMX485 sensor from STARVIS family of CMOS image sensors from Sony. The IMX485 is a 1/1.2" optical form-factor, CMOS image sensor with variable electronic shutter. e-CAM84_CUMI485C_MOD_H02R1 can stream video of 4K at 60 fps.

The front view of e-CAM84_CUMI485C_MOD_H02R1 is shown in the following figure.

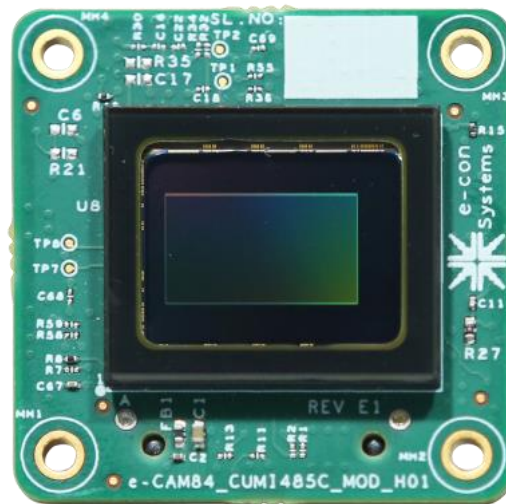


Figure 1: e-CAM84_CUMI485C_MOD_H02R1 Camera Module



4.1 Camera Module Features

The features of camera module are as follows:

- 1/1.2" optical form-factor, 8.4 MP camera module
- Manual focus/Auto focus lens *
- Low light camera sensor
- External ISP
- On-board micro-controller to communicate to sensor through I2C interface
- MIPI CSI-2 video output
- Capable of high frame rate video
- Small form factor enclosure with lens
- PCB Size : 30 mm x 30 mm
- Restriction of Hazardous Substances (RoHS) compliant
- One 26-pin SMT connectors
- Operating Voltage: 3.3V $\pm$ 5%, Power consumption: ~700mW

4.2 CMOS Image Sensor Features

The features of CMOS image sensor are as follows:

- IMX485 – 8.4 MP RAW 10/12-bit CMOS image sensor from SONY
- RGB RAW 10-bit/12-bit Data per pixel
- 1/1.2" optical form-factor
- Unit cell size: 2.9 μ m x 2.9 μ m

4.3 Maximum Frame Rate Supported

The maximum frame rate supported in e-CAM84_CUMI485C_MOD for 4-lane and 2-lane is listed in the table below.

Platform	Lanes	Resolution	Frame Rate(fps) in 10-bit Output	Frame Rate(fps) in 12-bit Output
NVIDIA® Jetson AGX ORIN™/Xavier™	4-lane	1920 x 1080*	90	90
		3840 x 2160	60	50
	2-lane	1920 x 1080*	62	62
		3840 x 2160	39	33

Table 1: Maximum Frame Rate Supported

*1944x1096 is cropped to 1920x1080 on the application end to skip the dummy bytes. 1944x1096 will be referred to as 1920x1080 throughout the document.

5 Pin Description

e-CAM84_CUMI485C_MOD_H02R1 has dual row, 26-pin connector. The pin numbers, signal names, pin types and their description from sensor perspective are listed in following table.



CN1 Pin No	Signal Name	Pin Type	Description
1	CAM_MIPI_CLK_N	OUTPUT	MIPI Clock Lane Differential Pair -
2	CAM_MIPI_DATA0_N	OUTPUT	MIPI Data Lane 0 Differential Pair -
3	CAM_MIPI_CLK_P	OUTPUT	MIPI Clock Lane Differential Pair +
4	CAM_MIPI_DATA0_P	OUTPUT	MIPI Data Lane 0 Differential Pair +
5	GND	POWER	Ground
6	GND	POWER	Ground
7	CAM_MIPI_DATA2_N	OUTPUT	MIPI Data Lane 2 Differential Pair -
8	CLK_I2C_SCL	Bi-Directional	I2C Clock
9	CAM_MIPI_DATA2_P	OUTPUT	MIPI Data Lane 2 Differential Pair +
10	I2C_SDA	Bi-Directional	I2C Data
11	GND	POWER	Ground
12	nRST_CONN	INPUT	Sensor reset
13	CAM_MIPI_DATA3_N	OUTPUT	MIPI Data Lane 3 Differential Pair -
14	CAM_PWDN_CONN	INPUT	Sensor Power Down
15	CAM_MIPI_DATA3_P	OUTPUT	MIPI Data Lane 3 Differential Pair +
16	GND	POWER	Ground
17	GND	POWER	Ground
18	HSYNC	INPUT	HSYNC
19	CAM_MIPI_DATA1_N	OUTPUT	MIPI Data Lane 1 Differential Pair -
20	VSYNC	INPUT	VSYNC
21	CAM_MIPI_DATA1_P	OUTPUT	MIPI Data Lane 1 Differential Pair +
22	XMASTER_CONN	INPUT	To BE FILLED BY SW TEAM
23	GND	POWER	Ground
24	TRIGGER	INPUT	To BE FILLED BY SW TEAM
25	FLASH	INPUT	To BE FILLED BY SW TEAM
26	3.3V	POWER	3.3V Supply

Table 2: Pin Descriptions

5.1 Mating Connector Details

The details of mating connectors are listed in following table.

Connector	Description	Manufacturer	Part Number
On-board connector	0.80 mm, Board - Board, 26 pin	Samtec	ERM8-013-03.0-L-DV-L-K-TR
Mating connector	0.80 mm, Board - Board, 26 pin	Samtec	ERF8-013-05.0-L-DV-L-K-TR

Table 3: Mating Connector Details

6 Electrical Specification

The electrical specification of e-CAM84_CUMI485C_MOD_H02R1 are as follows:

- [Recommended Operating Condition](#)
- [Functional Temperature Range](#)



- [DC Characteristics](#)
- [Power-Up Sequence](#)

6.1 Recommended Operating Condition

The recommended operating voltage and current consumption of e-CAM84_CUMI485C_MOD_H02R1 in active mode is listed in following table.

Platform	Lanes	Resolution	Frame Rate(fps) in 12-bit Output	Power Consumption(W) in 12-bit Output
NVIDIA® Jetson AGX ORIN™/Xavier™	4-lane	1920 x 1080	90	0.709
		3840 x 2160	50	1.138

Table 4: Recommended Operating Condition in Active Mode

6.2 Functional Temperature Range

The functional temperature range of e-CAM84_CUMI485C_MOD_H02R1 is listed in following table.

Temperature Range	Parameter Description
-30°C to 70°C	Electrically functional operating range

Table 5: Functional Temperature Range

Notes: -

- As temperature increases, the noise level also increases.
- Default lens (optional) supplied with this camera has operating temperature range of -20°C to 60°C. You can choose wider operating temperature lens as per your requirements.
- Performance guaranteed temperature range -10°C to 60°C

6.3 DC Characteristics

The typical condition: fEXTCLK = 37.125MHz; Analog Voltage (VDDHx, VDDSUB) $AV_{DD} = 2.9$ V; Interface Voltage (VDDMx) $OV_{DD} = 1.8$ V; Digital Voltage (VDDLx) $DV_{DD} = 1.2$ V; $T_J = 60^\circ\text{C}$ @ $T_A = 25^\circ\text{C}$.

The DC characteristics of e-CAM84_CUMI485C_MOD_H02R1 general purpose, Reset, Power Down are shown in the following table.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Digital Input Signals						
V_{IL}	Input voltage LOW	$V_{DD_IO} = 1.8$ V	-		$0.2 \times OV_{DD}$	V
V_{IH}	Input voltage HIGH	$V_{DD_IO} = 1.8$ V	$0.8 \times OV_{DD}$		-	V
Digital Output Signals						
V_{OL}	Output voltage LOW	$V_{DD_IO} = 1.8$ V	-		0.2	V
V_{OH}	Output voltage HIGH	$V_{DD_IO} = 1.8$ V	$OV_{DD} - 0.2$		-	V

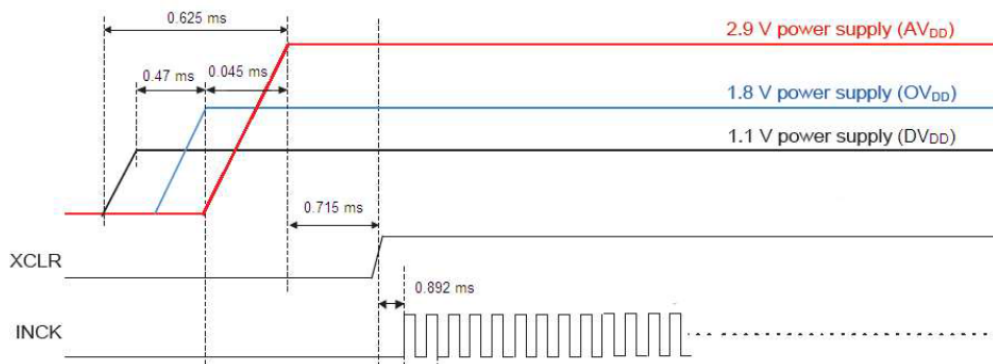


Table 6: DC Characteristics of IO Signals

Note: e-con Systems recommends the working voltage levels to be typically 1.8V_{DC} and not to reach the maximum limit.

6.4 Power-Up Sequence

The e-CAM84_CUMI485C_MOD_H02R1 camera module uses 1.1V for camera's digital core power which is generated from external 3.3V supply. The I²C activity must not be performed during power-up sequence. The power-up sequence recommended by e-con Systems in the customer design is shown in the following figure.

**Figure 2: Power-Up Sequence of e-CAM84_CUMI485C_MOD_H02R1**

The power-up sequence timing parameters are listed in the following table.

Symbol	Parameter	Minimum	Unit
t3	IO voltage to reset control delay	1	mS

Table 7: Power-Up Sequence Timing Parameters

Note: e-con Systems recommends implementing the power-up sequence as mentioned in this document.

7 Mechanical Specifications

The module size is 30 mm x 30 mm and the stack-up height of the board with its mating connector is 6 mm.

The e-CAM84_CUMI485C_MOD_H02R1 board drawings and dimensions are described in the following section.

7.1 e-CAM84_CUMI485C_MOD_H02R1 Mechanical Drawing

The top and bottom views of e-CAM84_CUMI485C_MOD_H02R1 mechanical drawing with optical orientations are shown in the following figures.



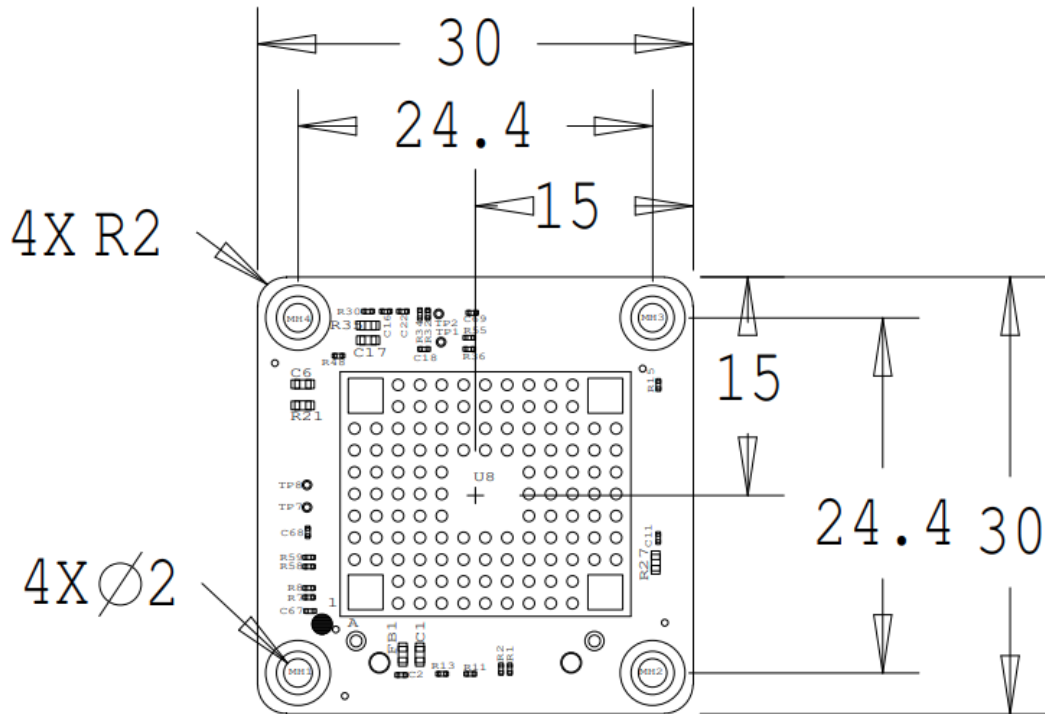


Figure 3: Top View of e-CAM84_CUMI485C_MOD_H02R1 Mechanical Drawing

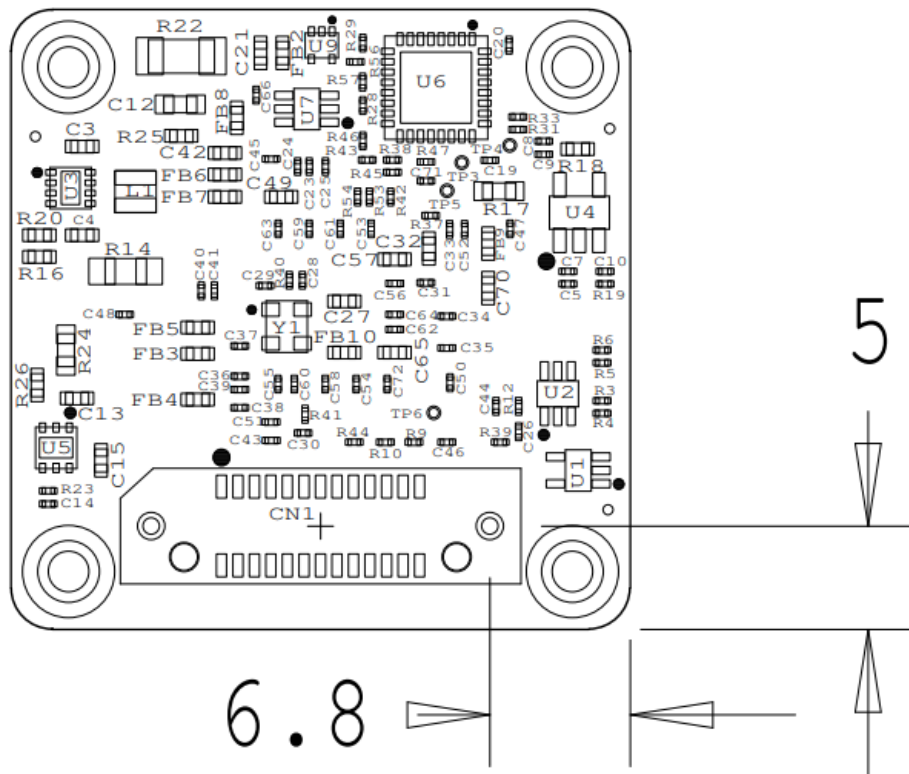


Figure 4: Bottom View of e-CAM84_CUMI485C_MOD_H02R1 Mechanical Drawing (Mirrored)



Note:

- All dimensions are in mm.
- Board outline dimension tolerance should be within $\pm 0.2\text{mm}$
- Further mechanical details will be provided based on your request.

7.2 Enclosure Drawing

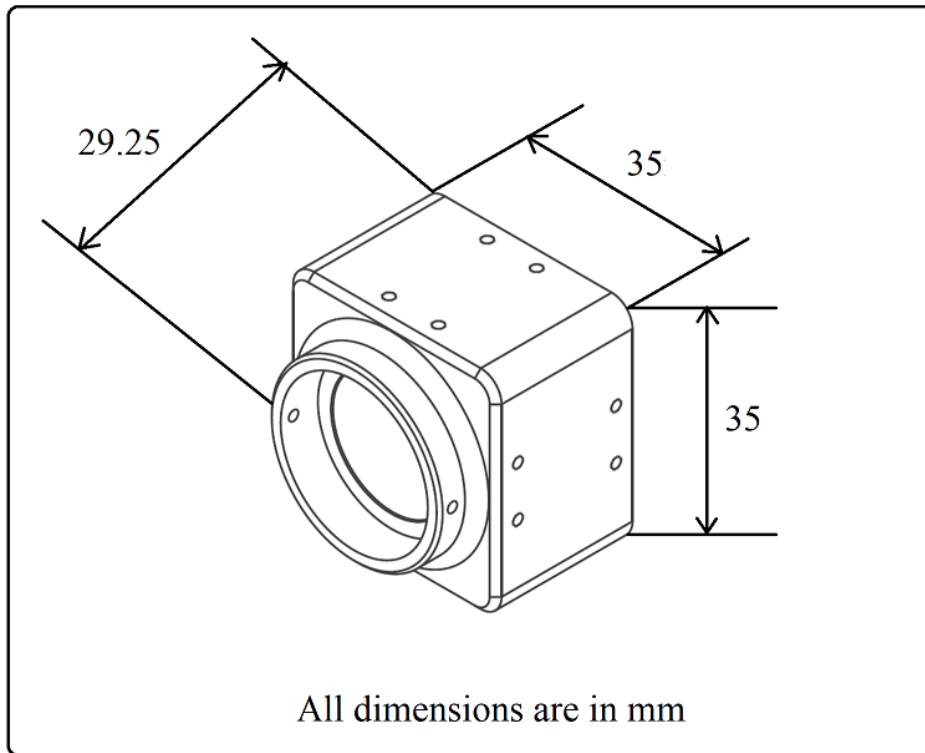


Figure.5: Enclosure Mechanical Diagram

Note: All dimensions are in mm.

7.3 Mechanical Part Details

The mechanical accessories for e-CAM84_CUMI485C_MOD_H02R1 camera board is listed in following table.

Part	Quantity	Specification
CS TO C Adapter Ring	1	Adapter ring for CS to C mount conversion

Table 8: Mechanical Part Details



Support

Contact Us

If you need any support on e-CAM84_CUMI485C_MOD_H02R1 product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

